

ORIGINAL ARTICLE

Concept Maps as a Content and Language Integrated Learning Facilitator in Digital Gaming Contexts: Triangulated Evidence From Students' Learning Achievements, Perceptions and Behavioural Patterns

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ABSTRACT

Background: While both Content and Language Integrated Learning (CLIL) and Digital Game-Based Learning (DGBL) are widely recognised as effective methods to enhance bilingual education, their combination often leads to cognitive overload and fragmented learning experiences. In Taiwan's bilingual education context, where integrating English into subject teaching is a national priority, finding effective instructional models remains a critical issue. To address this, the present study introduces the Concept Map-Facilitated Content and Language Integrated Digital Gaming (CM-CLDG) model, aiming to enhance bilingual science learning for elementary students by embedding concept mapping into a digital role-playing game.

Objective: Although DGBL has demonstrated potential to increase motivation and engagement, prior studies suggest that without structured guidance, learning outcomes may vary. The CM-CLDG model was designed to scaffold knowledge construction and reduce cognitive overload by integrating concept maps directly into the gaming tasks. This study aims to evaluate the effects of the CM-CLDG approach on students' English and science learning achievements, motivation, flow experience, anxiety and behavioural patterns.

Method: A quasi-experimental design was adopted with 78 sixth-grade students from an elementary school in central Taiwan. Participants were divided into the CM-CLDG group and the conventional CLIL gaming (C-CLDG) group. Pre- and posttests, questionnaires, behavioural log analyses and interviews were conducted to collect triangulated data. Learning behaviours were examined using sequential analysis via GSEQ.

Results and Conclusions: The CM-CLDG group significantly outperformed the C-CLDG group in both English and science achievements. Although no significant differences were found in motivation or anxiety, the C-CLDG group experienced greater flow. Behavioural analysis revealed that the CM-CLDG group exhibited more reflective, structured and strategic learning patterns. Interview findings supported these results, highlighting better memory retention and self-directed learning in the

1 | Introduction

Content and Language Integrated Learning (CLIL) has gained increasing attention in recent years as an effective instructional approach for promoting bilingual competence and subject learning simultaneously (Coyle et al. 2010; Jafarigohar et al. 2025). Based on the four Cs framework (Content, Communication, Cognition and Culture), CLIL has been widely adopted in European and Asian educational systems. In Taiwan, the government has promoted the 'Bilingual 2030' policy to integrate English into various subject areas in elementary and secondary schools. Despite its theoretical appeal, CLIL faces persistent challenges, including excessive cognitive load for learners, insufficient scaffolding for linking language with subject content and difficulties in sustaining engagement in cognitively demanding tasks (Dalton-Puffer 2011; Wong and Tian 2025). These limitations highlight the need for supportive pedagogical tools that can more effectively structure content-language integration.

In response to these issues, digital game-based learning (DGBL) has been proposed as a promising approach to support CLIL implementation in primary education. DGBL environments, characterised by interactive tasks, immediate feedback and immersive storylines, have been found to foster motivation and engagement among young learners (Gee 2003; van Eck 2006). Studies have reported the benefits of integrating computer games into language and content instruction, showing improvements in motivation, attention and participation (Bourgonjon et al. 2010; Inal and Cagiltay 2007). However, other researchers have cautioned that without proper instructional scaffolds, game-based activities may fail to produce significant learning gains (Charsky and Mims 2008; Fisch 2005). In particular, game-based tasks often lead to fragmented knowledge construction, as learners may focus on gameplay mechanics rather than building explicit connections between language and subject content. This indicates that DGBL needs to be complemented with structured tools to guide learners' cognitive processes and strengthen content-language integration.

Moreover, the arbitrary or poorly aligned use of learning strategies within games can undermine the learning experience. For instance, Charsky and Ressler (2011) observed that students required to use external tools for knowledge construction during gameplay exhibited lower motivation. These findings underscore the importance of embedding instructional strategies into game mechanics and narratives rather than treating them as add-ons. Among various instructional strategies, concept mapping is particularly noteworthy because it provides a visual and structural means for learners to organise complex relationships between language and content. Unlike other strategies that may feel peripheral to gameplay, concept maps can serve as an integral scaffold by externalising cognitive processes and making implicit knowledge connections explicit (Novak and Cañas 2008). Therefore, selecting concept mapping as the embedded strategy aims to address the problem of fragmented learning while minimising the risk of disengagement caused by unrelated external tools.

However, even when concept mapping strategies are embedded within the game narrative, their impact on learners' engagement and flow may vary. Hwang et al. (2013) found that while students using a concept map-embedded educational game showed significantly better learning achievement and lower cognitive load, their overall motivation and flow experience did not significantly differ from those who played a conventional game. One possible explanation is that concept mapping, despite aiding in knowledge organisation, may interrupt the immersive flow of gameplay due to its reflective and analytical nature. This aligns with Charsky and Ressler's (2011) argument that cognitive scaffolds, if overly effortful or externally imposed, can disrupt students' autonomy and enjoyment, ultimately reducing the affective benefits of game-based learning.

To address this issue, this study proposes the integration of concept mapping strategies directly into the structure of an educational role-playing game, as concept maps are well-regarded for supporting cognitive processing, aiding in the organisation of complex information and guiding metacognitive reflection (Novak and Cañas 2008). In a bilingual science learning context, concept maps can serve both as content organisers and language scaffolds, supporting the dual goals of CLIL instruction. By embedding concept map construction tasks within gaming missions, learners can contextualise vocabulary, reinforce scientific concepts and articulate their understanding within the game environment.

An experiment was conducted based on a bilingual science learning unit on plant knowledge at an elementary school. During gameplay, students were asked to collect content-related information and complete concept maps as part of mission objectives. The effectiveness of this concept map-embedded game-based learning approach was evaluated in comparison with a conventional educational game setting. The following research questions were examined:

1. Do the students who learn with the concept map-facilitated content and language integrated digital gaming (CM-CLDG) model show better English learning achievements than those who learn with the conventional content and language integrated digital gaming (C-CLDG) model?
2. Do the students who learn with the CM-CLDG model show better science learning achievements than those who learn with the C-CLDG model?
3. Do the students who learn with the CM-CLDG model show higher learning motivation than those who learn with the C-CLDG model?
4. Do the students who learn with the CM-CLDG model experience greater flow than those who learn with the C-CLDG model?
5. Does the CM-CLDG model reduce students' learning anxiety compared to the C-CLDG model?

Summary

- What is already known about this topic
 - Content and Language Integrated Learning (CLIL) aims to enhance subject knowledge and language proficiency simultaneously.
 - Digital Game-Based Learning (DGBL) has shown potential to motivate young learners and foster engagement.
 - The integration of instructional scaffolds into game mechanics can enhance learning but may also interfere with immersion if not well-designed.
- What this paper adds
 - A concept map-facilitated digital gaming model (CM-CLDG) is proposed to address the cognitive load and learning structure challenges in CLIL-based game environments.
 - The CM-CLDG model embeds concept mapping tasks into a bilingual science game to support both English and science learning in elementary education.
 - Empirical results show that the CM-CLDG approach significantly improves learning achievement and supports strategic, reflective behavioural patterns compared to conventional CLIL gaming (C-CLDG).
- Implications for practice and/or policy
 - The CM-CLDG approach can help educators design structured yet engaging bilingual learning environments, particularly in elementary science and language instruction.
 - Embedding concept maps into digital games promotes deeper understanding and self-directed learning without compromising engagement.
 - It is recommended that teachers apply the CM-CLDG approach to other bilingual education courses for EFL learners.

6. Do the students who learn with the CM-CLDG model exhibit different behavioural patterns than those who learn with the C-CLDG model?

2 | Literature Review

2.1 | DGBL

DGBL is an educational approach that integrates game mechanics and learning objectives to engage students in meaningful learning tasks (Hwang et al. 2020; Panjaburee et al. 2025; Prensky 2007). In DGBL environments, learners interact with computer-based games where educational content is embedded within gaming scenarios, guiding learners to accomplish tasks through exploration, decision-making and problem-solving (Poompimol et al. 2024; Prensky 2007). These games are often designed to create immersive learning experiences by offering challenges that foster cognitive development and skill acquisition, with the support of instant feedback mechanisms that promote reflection and revision (Kinzie and Joseph 2008; Oblinger 2004). Learners are typically required to complete missions or solve problems that align with educational goals,

encouraging both enjoyment and deep learning (Cagiltay 2007; Collier and Scott 2009). Due to its interactive and engaging nature, DGBL has been applied in various educational contexts such as mathematics, language learning, science and vocational training (Liu and Chu 2010).

In recent years, many researchers have explored the effectiveness of DGBL in terms of enhancing different aspects of student learning, including learning achievement, motivation, flow experience, anxiety and learning behaviour. For instance, Hwang et al. (2020) conducted a meta-analysis on science, technology, engineering and mathematics (STEM) education and found that DGBL had a significant positive impact on students' academic achievement. Other studies (e.g., Chen and Hwang 2020; Hwang and Zou 2020) have shown that DGBL enhances students' learning motivation through personalised or gamified designs. The use of digital games has also been associated with improved flow experience, where students become deeply engaged in tasks. Furthermore, DGBL has demonstrated potential in reducing learning anxiety, especially when incorporating adaptive feedback or agent-based systems. In terms of behaviour, recent studies have analysed learning logs and behaviour patterns to evaluate how students interact with game elements, make decisions and adjust strategies during gameplay.

However, despite its advantages, DGBL also faces several challenges in educational practice. One concern is the alignment of game mechanics with pedagogical objectives. Without careful integration, learners may focus more on winning the game rather than on acquiring knowledge (Charsky and Ressler 2011). Moreover, game design that lacks adequate scaffolding or personalised support may result in increased cognitive load or disengagement (Ke 2008). Additionally, there are issues related to assessment, as traditional evaluation methods may not fully capture the depth of learning occurring in game-based environments (Nelson et al. 2011). As such, it is crucial to embed effective instructional strategies and assessment tools within DGBL environments to ensure that educational goals are met while maintaining the motivational appeal of games.

2.2 | Concept Maps

Concept maps are visual learning tools that enable learners to represent, organise and connect their understanding of knowledge domains through diagrams that illustrate the relationships among concepts (Novak and Gowin 1984; Novak et al. 1983). By organising concepts hierarchically and showing linkages between them, concept maps help learners clarify complex ideas, identify misconceptions and support meaningful learning (Chiou 2008). In instructional contexts, concept maps have been widely used as tools for knowledge construction (Hwang and Zou 2020), learning diagnosis (Chang et al. 2005) and formative assessment (Cai et al. 2025). Their application has spanned multiple disciplines, including science, language education and computer programming.

Recent studies have investigated the effectiveness of concept maps in terms of supporting various learning outcomes, with

multiple studies showing significant benefits for learning achievement (Chang et al. 2022; Huang et al. 2022; Hwang and Zou 2020; Kienitz et al. 2025). For example, Chen and Hwang (2020) found that a concept mapping-based flipped classroom improved English learners' speaking performance. As for learning motivation, the visual and integrative nature of concept mapping has been found to enhance learners' engagement and perceived usefulness of tasks (Novak and Cañas 2008; Sun and Lee 2016). Moreover, Huang et al. (2022) indicated that students using concept mapping strategies in science inquiry displayed higher levels of motivation and participation. Regarding learning anxiety, research is relatively limited. However, Chen and Hwang (2020) reported that students in concept mapping-based environments showed reduced speaking anxiety. While fewer studies have directly linked concept maps to reductions in learning anxiety, the structure and clarity they provide may indirectly mitigate confusion-related stress. In terms of learning behaviour, studies have applied behaviour sequence analysis and learning log interpretation to understand how students engage with concept maps. Kienitz et al. (2025) and Hwang and Zou (2020) found that learners using concept maps exhibited more strategic and reflective learning behaviours. Chang et al. (2022) further summarised that concept maps help scaffold self-regulated learning behaviours across K-12 settings. It is worth noting that while concept maps have been widely used to support various learning outcomes, research directly linking concept maps to flow experience is currently limited. This represents a gap in the literature. Given that flow experience is characterised by deep engagement and immersion, future studies could explore how concept mapping strategies, especially when integrated with interactive environments like games, may facilitate flow among learners.

While prior studies have respectively examined the benefits of DGBL and concept mapping, research that integrates these two approaches remains scarce, not to mention in CLIL contexts. DGBL has been widely recognised for enhancing motivation and engagement, yet it often results in fragmented knowledge construction and insufficient scaffolding for integrating subject content during the gaming process (e.g., Zhao et al. 2025). Without explicit structural support, learners may struggle to connect in-game experiences with underlying disciplinary concepts, leading to surface-level or disjointed understanding. Concept mapping, by contrast, provides an evidence-based means of externalising cognitive processes, reducing extraneous cognitive load and strengthening coherence among concepts (Cheng et al. 2024; Chu et al. 2025). Prior research has demonstrated its effectiveness in terms of promoting meaningful learning by making relationships between ideas visible and facilitating reflection (Chang et al. 2016). Embedding concept mapping into DGBL, therefore, offers a theoretically grounded solution to address the shortcomings of game-based environments. Specifically, it can scaffold learners in constructing integrative knowledge, support bilingual content-language connections and guide them toward more reflective and strategic learning behaviours. This rationale directly underpins the design of the CM-CLDG model.

Building on Cognitive Load Theory (Sweller 1988, 1994), which emphasises the importance of reducing extraneous

load and fostering germane load for effective learning, this study proposes a CM-CLDG model. This model integrates concept mapping as an embedded component of the digital gaming process, allowing learners to visualise relationships among concepts as they progress through gaming tasks. The CM-CLDG model aims to leverage the cognitive benefits of concept maps and the motivational affordances of DGBL, supporting students in constructing knowledge, managing emotions and engaging meaningfully with learning content. By embedding concept mapping into the gaming storyline and feedback mechanisms, this approach seeks to enhance learning outcomes in a more structured and metacognitively supportive way.

3 | Concept Map-Facilitated Content and Language-Integrated Digital Gaming Model

3.1 | Framework and System Structure

In this study, a concept map-facilitated content and language-integrated digital gaming model is proposed by embedding concept mapping strategies into gaming scenarios to facilitate students' learning performance in an integrated English and science course for EFL (English as a Foreign Language) students. The digital game, 'SF Queen's Adventure', was implemented using the RPG Maker platform developed by Enterbrain Incorporation. Figure 1 illustrates the structure of the concept map-embedded digital game, including a gaming module, a concept mapping module, a gaming rule database, a gaming depository database, a learning materials database and a set of pre-defined concept maps. Throughout gameplay, students interact with the Queen character, who provides guidance and assigns missions to collect badges within various habitat environments. Their learning portfolios, including completed and uncompleted missions, badges collected during the exploration of habitats, and responses to embedded quiz questions, are systematically recorded in the gaming depository database. Subsequently, the concept mapping module assists learners in organising and structuring the collected data according to concept map templates designed and provided by instructors, enabling them to effectively integrate knowledge and language skills within the game's narrative context.

3.2 | Gaming Flow and Interface

This digital game system is designed around a learning unit focused on animal habitats and their associated environmental characteristics. The game is structured around a series of missions that guide students through a sequenced process of exploration, interaction and knowledge construction. As shown in Figure 2, the gameplay begins with the initiation of the game and presentation of scene-specific information. Students then answer quiz questions based on the habitat content. Based on their responses, the system provides immediate feedback indicating whether their answers are correct or incorrect, followed by the opportunity to collect badges as a reward. During the gaming process, the system also includes opportunities for students to retry missions, identify correct and incorrect information and receive final explanations. In the later stages, students

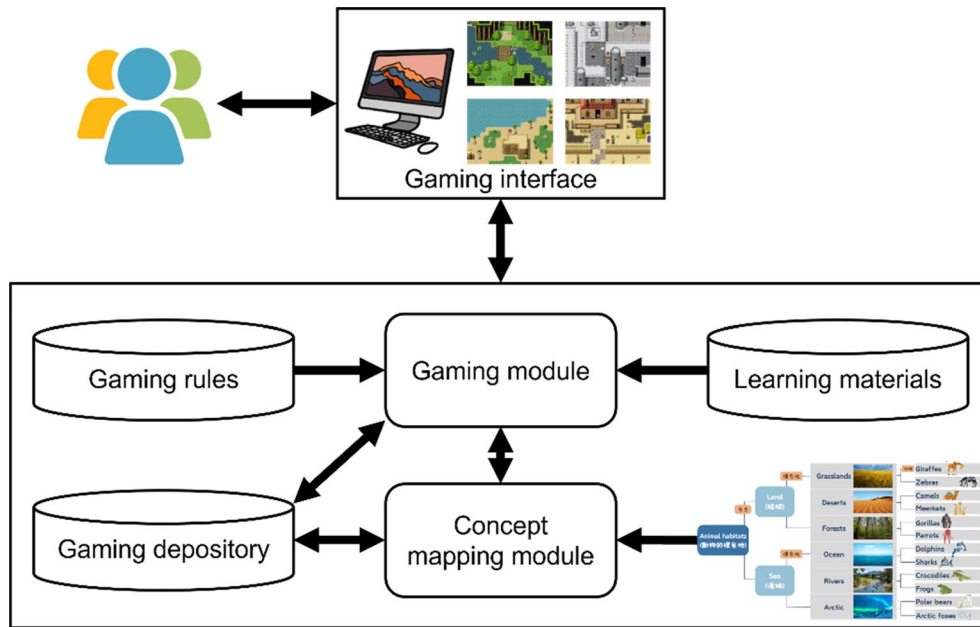


FIGURE 1 | Structure of the concept map-embedded digital game.

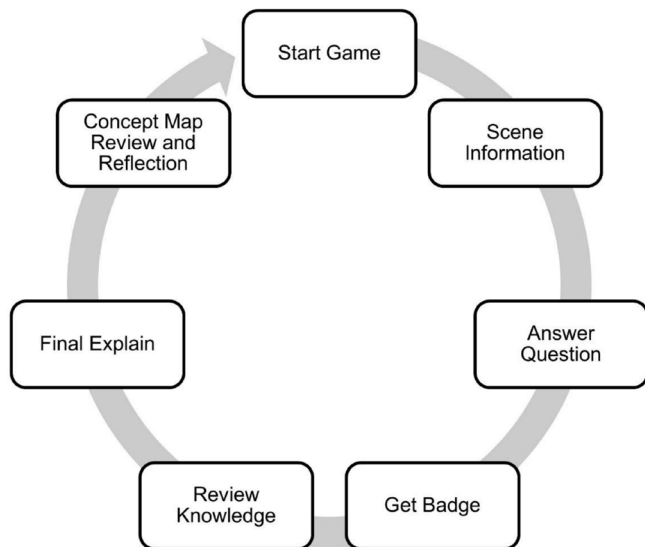


FIGURE 2 | Task design flow of the digital game.

review what they have learned through system feedback and then engage in a concept map review and reflection stage, where they view and organise the integrated concept map to consolidate their understanding. This step closes the loop by linking task execution back to reflection, thereby forming a cyclical learning process that promotes both formative assessment and meaningful learning.

The game incorporates multiple immersive scenarios, designed to engage students deeply with tasks related to different animal habitats. Figure 3 shows a narrative interaction scenario, in which students converse with the Queen character to receive background information and begin their mission. This dialogue-based setup helps establish the storyline and contextualise the learning objectives. Figure 4 presents another interactive scenario where students engage with a learning target (e.g., the



FIGURE 3 | Gaming scenario of completing the mission.

Frog King) to answer habitat-related questions such as, 'Where do crocodiles live?' Through these interactions, students actively collect and apply knowledge across diverse environments such as grasslands, deserts, forests, oceans, rivers and the arctic. The collected information is stored in the system for later review and integration.

To facilitate structured knowledge integration, the game utilises embedded concept maps that guide students in organising and applying collected information. Figure 5 demonstrates a concept map template integrated into a mission task, in which students complete the diagram by matching animals to their corresponding habitats through multiple-choice selections. This visual format encourages meaningful connections between species and environments. Figure 6 illustrates the final concept map displayed before game completion, summarising all habitat-animal relationships learned throughout gameplay. This concluding visualisation reinforces students' understanding and supports reflective learning.



FIGURE 4 | Gaming scenario of collecting information about animal habitats.

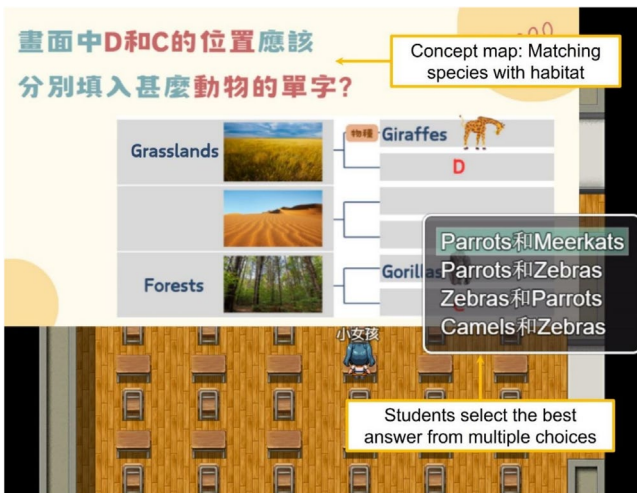


FIGURE 5 | Illustrative example of a concept map template embedded in a gaming mission.

4 | Experimental Design

To evaluate the effectiveness of the proposed CM-CLDG approach, an experiment was conducted in an English course focusing on the topic of animal habitats at an elementary school. This unit was integrated into the selected school's English curriculum and was designed to support the national 'Bilingual 2030' initiative in Taiwan. The aim of the study was to examine the effects of the proposed approach on students' English and science learning achievements, learning motivation, flow experience, learning anxiety and their behavioural patterns.

4.1 | Participants

A quasi-experiment was conducted in this study. The participants were three classes of sixth-grade students from an elementary school in central Taiwan. Their average age was 12 years old. They were divided into two groups: the CM-CLDG model group, consisting of two classes of students ($n=53$), and the C-CLDG model group, consisting of one class ($n=25$). Both groups were taught by the same instructor, who had over 3 years of English teaching experience. In addition, a senior teacher with

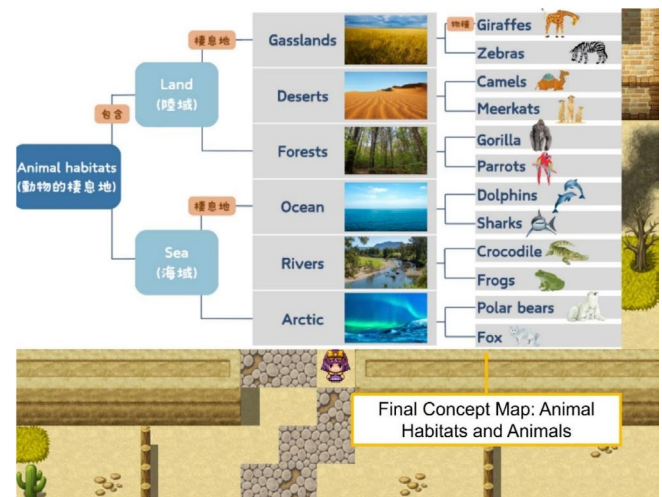


FIGURE 6 | Integrated concept map displayed before game completion.

more than 8 years of experience in bilingual science instruction participated in the course design and discussion to ensure the pedagogical quality and alignment with both English and science education goals.

4.2 | Research Tools

The research tools used in this study included pre- and posttests for English and science achievement, questionnaires for measuring learning motivation, flow experience and learning anxiety, as well as a behaviour coding scheme.

The English achievement pre- and posttest were developed by three experienced teachers. The pretest aimed to assess students' prior English knowledge, focusing on vocabulary and sentence comprehension related to animals and daily conversations. It consisted of vocabulary recognition, sentence-picture matching and reading comprehension sections, with a perfect score of 100. The posttest, in contrast, focused on animal habitats and species, including vocabulary identification, species-habitat matching and a cloze test. The perfect score of the posttest was 100.

Similarly, the science achievement pre- and posttest were developed by three experienced teachers. The pretest aimed to assess students' prior knowledge of local ecosystems, focusing on animals and their living environments around their residential area. The posttest covered the target content of the learning unit, focusing on six different animal habitats. Both the pre- and posttest consisted of 25 multiple-choice items, with a perfect score of 100.

The learning motivation questionnaire was adapted from the instrument developed by Yin et al. (2021), consisting of six items rated on a 5-point Likert scale. Sample items include 'I enjoy challenging materials in the course because I can learn new things' and 'I feel most satisfied when I achieve good grades in the course'. The Cronbach's alpha value of the questionnaire was 0.79, indicating acceptable reliability.

The flow experience questionnaire was adapted from the instrument developed by Pearce et al. (2005), consisting of nine items rated on a 5-point Likert scale. Sample items include 'I feel confident about what I am doing in this course, and the outcomes meet my expectations' and 'I feel that time passes quickly during the course'. The Cronbach's alpha value of the questionnaire was 0.90, indicating high reliability.

The learning anxiety questionnaire was adapted from the instrument developed by Yin (2011), consisting of six items rated on a 5-point Likert scale. Sample items include 'I usually do not feel worried when learning English' and 'I feel relaxed and comfortable during English lessons'. The Cronbach's alpha value of the questionnaire was 0.90, indicating high reliability.

Additionally, a behaviour coding scheme was developed to analyse students' learning behaviours during gameplay. The scheme included codes for key behaviours such as starting the game, reviewing scene information (SI), answering questions, obtaining badges and constructing or sharing knowledge, as shown in Table 1. All gameplay sessions were recorded for further analysis. The coding was independently conducted by two experienced coders. To ensure reliability, the coding results were compared, and any discrepancies were resolved through discussion until consensus was reached. This process helped ensure consistency and accuracy in the coding results.

4.3 | Experiment Procedures

Figure 7 shows the procedure of the experiment. The participants were divided into two groups: the CM-CLDG model group (experimental group) and the C-CLDG model group (control group).

At the beginning of the experimental session, all students took the English and science pretests, as well as completed the questionnaires measuring learning motivation, flow experience and learning anxiety. During the intervention, the CM-CLDG model group learned with the CM-CLDG model, while the C-CLDG model group learned with the C-CLDG model. Both versions of the game included the same storyline, missions and learning content, ensuring strict control of variables such as instructional materials, tasks and duration. The critical distinction was that the CM-CLDG group was required to complete embedded concept mapping tasks, in which students organised key vocabulary, sentence patterns and scientific concepts into instructor-provided concept map templates while progressing through the game. In contrast, the C-CLDG group proceeded with the same game tasks but without any concept mapping activity. It should be noted that although the total experimental duration amounted to 200 min, the sessions were not conducted continuously on the same day. Each session lasted around 40 min and was interspersed with sufficient breaks, ensuring that students could maintain attention and reduce fatigue. This arrangement was designed to fit the typical concentration span of elementary school students and to minimise the potential influence of fatigue on their posttest performance. The students were given 120 min to complete the game and were allowed to replay it within the given time.

TABLE 1 | Coding scheme.

Code	Phase	Description
SG	Start Game	The student enters the game and begins interacting with the NPC.
SI	Scene Info	The student reads contextual information about the task or game scene.
AQ	Answer Question	The student attempts to answer the question presented in the task.
CA	Correct Answer	The student answers correctly and receives positive feedback.
WA	Wrong Answer	The student answers incorrectly and receives negative feedback.
GB	Get Badge	The student receives a badge as a reward after completing the task.
RK	Review Knowledge	The student reviews previously acquired knowledge.
FE	Final Explain	The student listens to the NPC's summary or final remarks for the task.
FC	First Clear	The student clears all tasks and collects all badges on the first attempt.
RC	Retry Clear	The student clears all tasks and collects all badges after retrying.
CR	Catch Right	The student correctly catches the designated target (e.g., animal).
CW	Catch Wrong	The student incorrectly catches a nontarget item (e.g., wrong animal).

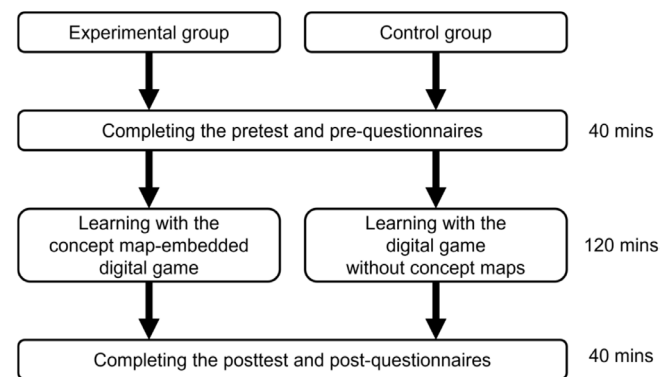


FIGURE 7 | Procedure of the experiment.

TABLE 2 | ANCOVA result of the English learning achievement posttest of the two groups.

Group	N	Mean	SD	Adjusted mean	Std error	F	η^2
CM-CLDG	53	95.32	12.55	94.59	1.90	30.01***	0.286
C-CLDG	25	74.56	23.55	76.12	2.77		

*** $p < 0.001$.**TABLE 3** | ANCOVA result of the science learning achievement posttest of the two groups.

Group	N	Mean	SD	Adjusted mean	Std error	F	η^2
CM-CLDG	53	85.21	8.32	84.89	1.52	27.68***	0.270
C-CLDG	25	70.08	19.05	70.76	2.21		

*** $p < 0.001$.

After the learning activity, all students completed the posttests for English and science, as well as the questionnaires measuring learning motivation, flow experience and learning anxiety. In addition, the behaviour logs during gameplay were collected and analysed to compare the behavioural patterns of the two groups.

4.4 | Data Analysis

The sample sizes of the experimental group ($n = 53$) and the control group ($n = 25$) were unequal. Nonetheless, prior research has shown that both ANCOVA and t tests are robust to such discrepancies, particularly when the assumption of homogeneity of variance is met (Field 2013; Tabachnick and Fidell 2019). In the present study, Levene's test confirmed that the homogeneity of variance assumption was satisfied ($p > 0.05$), ensuring the appropriateness of ANCOVA for subsequent analysis. It should also be noted that this study employed a quasi-experimental design, in which all participants were from intact, pre-existing classes as originally grouped by the school. No reassignment or regrouping of students was conducted, and all lessons were delivered by the original classroom teachers as part of the formal curriculum schedule. Therefore, the data structure did not involve hierarchical clustering at the class level, and the use of ANCOVA was appropriate for addressing the research questions. While Tsao et al. (2025) also conducted a study with unequal group sizes in a game-based learning context and adopted ANCOVA for analysis, their group ratio was different from that of the present study. Thus, their work is cited only as a supplementary reference to illustrate that ANCOVA has been commonly used in similar educational settings, rather than as direct evidence of equivalence. Taken together, these considerations suggest that the statistical analyses in the present study were valid, and the results can be interpreted with confidence.

5 | Results

5.1 | English Learning Achievement

The one-way analysis of covariance (ANCOVA) was employed using the pretest scores of English learning achievement as the covariate, the learning approach (concept map-facilitated or conventional gaming) as the independent variable, and the posttest scores of English learning achievement as the dependent variable.

Before conducting the ANCOVA, the assumption of homogeneity of regression slopes was tested. The result showed no significant violation ($F = 0.45$, $p > 0.05$), indicating that the two groups had equivalent prior knowledge before the learning activity.

As shown in Table 2, the adjusted mean score of the CM-CLDG group was 94.59 ($SE = 1.90$), while that of the C-CLDG group was 76.12 ($SE = 2.77$). After excluding the effect of the pretest scores, there was a significant difference between the two groups ($F(1, 75) = 30.01$, $p < 0.001$), with a large effect size ($\eta^2 = 0.286$). This indicates that students in the concept map-facilitated group outperformed those in the conventional group in terms of English learning achievement, suggesting that embedding concept mapping into digital gameplay can significantly enhance students' language learning performance.

5.2 | Science Learning Achievement

The ANCOVA was employed using the pretest scores of science learning achievement as the covariate, the learning approach (concept map-facilitated or conventional gaming) as the independent variable, and the posttest scores of science learning achievement as the dependent variable.

Before conducting the ANCOVA, the assumption of homogeneity of regression slopes was tested. The results showed no significant violation ($F = 2.76$, $p > 0.05$), indicating that the two groups had equivalent prior knowledge before the learning activity.

As shown in Table 3, the adjusted mean score of the CM-CLDG group was 84.89 ($SE = 1.52$), while that of the C-CLDG group was 70.76 ($SE = 2.21$). After excluding the effect of the pretest scores, there was a significant difference between the two groups ($F(1, 75) = 27.68$, $p < 0.001$), with a large effect size ($\eta^2 = 0.270$). This suggests that concept map-facilitated gaming learning was also effective in terms of enhancing students' science learning outcomes.

5.3 | Learning Motivation

The ANCOVA was employed using the prequestionnaire scores of learning motivation as the covariate, the learning approach (concept map-facilitated or conventional gaming) as the

TABLE 4 | ANCOVA result of the learning motivation posttest of the two groups.

Group	N	Mean	SD	Adjusted mean	Std error	F	η^2
CM-CLDG	53	3.71	0.69	3.77	0.47	0.02	0.000
C-CLDG	25	3.90	0.66	3.79	0.69		

TABLE 5 | The *t* test result of the flow experience scores of the two groups.

Group	N	Mean	SD	<i>t</i>	<i>d</i>
CM-CLDG	53	4.00	0.83	-2.59*	0.58
C-CLDG	25	4.35	0.68		

* $p < 0.05$.

independent variable, and the postquestionnaire scores of learning motivation as the dependent variable.

Before conducting the ANCOVA, the assumption of homogeneity of regression slopes was tested. The result showed no significant violation ($F = 0.68$, $p > 0.05$), indicating that the two groups had equivalent levels of learning motivation prior to the activity.

As shown in Table 4, the adjusted mean score of the CM-CLDG group was 3.77, while that of the C-CLDG group was 3.79. After excluding the effect of the prequestionnaire scores, no significant difference was found between the two groups ($F = 0.02$, $p > 0.05$), with a negligible effect size ($\eta^2 = 0.000$). This indicates that both groups maintained similar levels of learning motivation after the learning activity, regardless of whether concept maps were incorporated into the digital gameplay.

5.4 | Flow Experience

To examine the impact of different learning approaches on students' flow experience during DGBL, the independent samples *t* test was conducted. As shown in Table 5, the mean and standard deviation of the flow experience scores of the CM-CLDG group were 4.00 and 0.83, respectively, while those of the C-CLDG group were 4.35 and 0.68.

The *t* test result ($t(76) = -2.59$, $p < 0.05$, $d = 0.58$) indicates a significant difference between the two groups. According to Cohen (1988), a *d* value around 0.5 suggests a medium effect size. This implies that students in the conventional gaming group reported a significantly higher sense of immersion compared to those in the concept map-facilitated group.

5.5 | Learning Anxiety

The one-way ANCOVA was employed using the prequestionnaire scores of learning anxiety as the covariate, the learning approach (concept map-facilitated or conventional gaming) as the independent variable and the postquestionnaire scores of learning anxiety as the dependent variable.

Before conducting the ANCOVA, the assumption of homogeneity of regression slopes was tested. The result showed no significant violation ($F = 0.49$, $p > 0.05$), indicating that the two groups had comparable levels of learning anxiety prior to the activity.

As shown in Table 6, the adjusted mean score of the CM-CLDG group was 3.77, while that of the C-CLDG group was 3.66. After controlling for the prequestionnaire scores, the ANCOVA results revealed no significant difference between the two groups ($F = 0.36$, $p > 0.05$), with a very small effect size ($\eta^2 = 0.005$). This suggests that the integration of concept maps into the digital gaming environment did not have a notable influence on students' learning anxiety levels.

5.6 | Learning Behaviours

During the learning activities, all student behaviours were automatically recorded by the digital learning system. Sequential analysis using Generalised Sequential Quierier (GSEQ) was conducted to explore the behavioural patterns of students under the two different conditions: the CM-CLDG model and the C-CLDG model. A total of 53 students participated in the CM-CLDG model, producing 4449 behaviours, whereas 25 students engaged in the C-CLDG model, generating 1638 behaviours.

The results of the sequential analysis for the CM-CLDG model are presented in Table 7, while those for the C-CLDG model are presented in Table 8. Sequential analysis calculates Z-scores to determine statistically significant transitions between behaviours. According to Bakeman and Gottman (1997) and Bakeman and Quera (2011), a Z-score greater than 1.96 indicates a statistically significant sequential relationship between two behaviours. For instance, in the CM-CLDG model, a significant behavioural path from behaviour AQ (Answer Question) to behaviour CA (Correct Answer) was observed, highlighting an effective pattern of active engagement leading to successful task completion.

Figure 8 illustrates the behavioural transitions of the CM-CLDG group. It shows structured and reflective learning patterns characterised by sequences such as AQ directly transitioning to CA, indicating efficient engagement and understanding. Additionally, sequences demonstrating effective error correction and refinement strategies were evident, including CW (Catch Wrong) transitioning to CR (Catch Right). The presence of iterative cycles among certain behaviours, such as CR transitioning repeatedly to itself, and subsequently to FC (First Clear), suggests sustained strategic engagement, reflective thinking and thorough knowledge review processes within this group.

Figure 9 represents the behavioural transitions of the C-CLDG group. Although beneficial sequences such as AQ leading to CA and CW transitioning to CR were also present, the overall pattern

TABLE 6 | ANCOVA result of the learning anxiety posttest of the two groups.

Group	N	Mean	SD	Adjusted mean	Std error	F	η^2
CM-CLDG	53	3.66	1.03	3.77	0.60	0.36	0.005
C-CLDG	25	3.90	0.87	3.66	0.81		

TABLE 7 | Results of sequential analysis of behaviours in the CM-CLDG model.

Z-score	Target											
Given	SG	SI	AQ	CA	WA	GB	RK	FE	FC	RC	CR	CW
SG	0	18.16*	-4.36	-3.06	-2.1	-1.41	-3.8	-0.71	-0.47	-0.52	-1.29	-0.73
SI	0	-9.6	12.71*	-11.22	-7.72	-5.17	10.19*	18.16*	-1.74	-1.9	-4.73	-2.67
AQ	0	-16.01	-23.22	46.77*	32.19*	-8.61	-16.29	-4.36	-2.9	7.93*	-7.88	-4.45
CA	0	18.05*	0.91	-13.11	-9.03	21.01*	-2.83	-3.06	-2.04	-2.22	-5.52	-3.12
WA	0	-7.72	32.19*	-9.03	-6.21	-4.16	-11.23	-2.1	-1.4	-1.53	-3.8	-2.15
GB	0	35.88*	-8.61	-6.04	-4.16	-2.78	-7.51	-1.41	-0.94	-1.02	-2.54	-1.44
RK	0	-3.81	1.37	-16.32	-11.23	0.99	28.48*	-3.8	-2.53	-2.77	-6.87	-3.88
FE	0	-2.59	-4.31	-3.02	-2.08	-1.39	-3.76	-0.7	-0.47	-0.51	22.54*	23.42*
FC	0	0	0	0	0	0	0	0	0	0	0	0
RC	0	0	0	0	0	0	0	0	0	0	0	0
CR	0	-4.73	-4.57	-5.52	-3.8	-2.54	-6.87	-1.29	22.17*	-0.94	37.25*	20.17*
CW	0	-2.67	-2.23	-3.12	-2.15	-1.44	-3.88	-0.73	3.71*	-0.53	26.54*	6.14*

*Z > 1.96.

TABLE 8 | Results of sequential analysis of behaviours in the C-CLDG model.

Z-score	Target											
Given	SI	AQ	CA	WA	GB	RK	FE	FC	RC	CR	CW	SI
SG	0	11.5*	-2.8	-1.99	-1.29	-0.89	-2.66	-0.53	-0.28	-0.44	-0.84	-0.64
SI	0	1.15	1.57	-7.02	-4.52	-3.14	6.83*	11.5*	-0.98	-1.56	-2.94	-2.25
AQ	0	-9.87	-12.32	28.79*	18.55*	-4.76	-10.34	-2.8	-1.49	6.4*	-4.46	-3.42
CA	0	6.33*	2.96*	-7.57	-4.88	10.45*	-0.22	-1.99	-1.06	-1.68	-3.17	-2.43
WA	0	-4.52	17.26*	-4.88	-3.14	-2.18	-5.18	-1.29	-0.68	-1.08	-2.04	-1.57
GB	0	19.5*	-4.76	-3.38	-2.18	-1.51	-4.51	-0.89	-0.47	-0.75	-1.42	-1.09
RK	0	-2.43	3.04*	-10.1	-6.51	1.92	14.49*	-2.66	-1.42	-2.24	-4.23	-3.24
FE	0	-1.85	-2.31	-1.99	-1.29	-0.89	-2.66	-0.53	-0.28	-0.44	14.2*	12.25*
FC	0	0	0	0	0	0	0	0	0	0	0	0
RC	0	0	0	0	0	0	0	0	0	0	0	0
CR	0	-2.94	-1.92	-3.17	-2.04	-1.42	-4.23	-0.84	11.22*	-0.7	17.16*	15.53*
CW	0	-2.25	-0.97	-2.43	-1.57	-1.09	-3.24	-0.64	2.66*	-0.54	16.56*	8.53*

*Z > 1.96.

of behavioural transitions was less clearly structured. Notably, recurrent bidirectional transitions occurred frequently, such as CA reverting back to AQ, indicating potential uncertainties or inconsistent comprehension during tasks. Moreover, repetitive

loops within single behaviours (e.g., CW repeatedly transitioning to itself) reflect difficulties in effectively addressing mistakes or improving understanding. Such behavioural patterns indicate relatively less strategic orientation and less systematic

approaches toward learning compared to those evident in the CM-CLDG group. Overall, the sequential analysis highlights clear qualitative differences between behavioural patterns in the two gaming models.

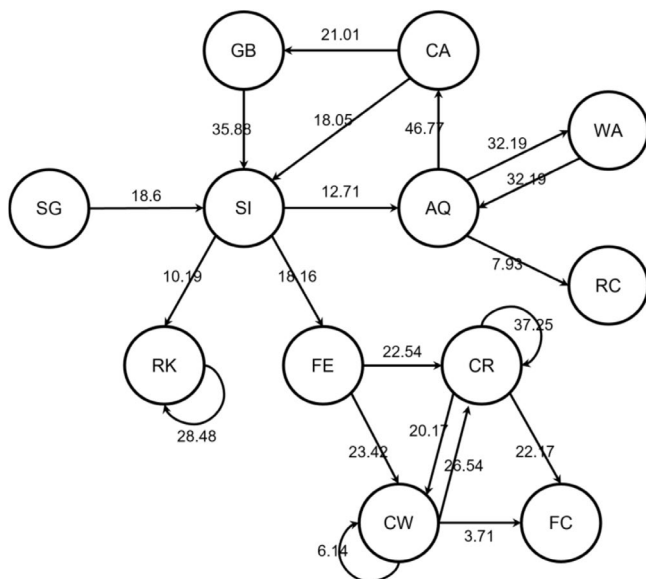


FIGURE 8 | Behavioural sequence diagram of the CM-CLDG group.

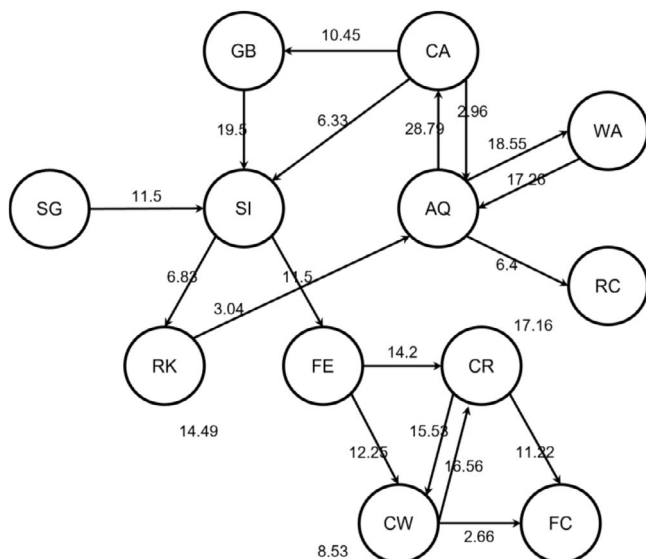


FIGURE 9 | Behavioural sequence diagram of the C-CLDG group.

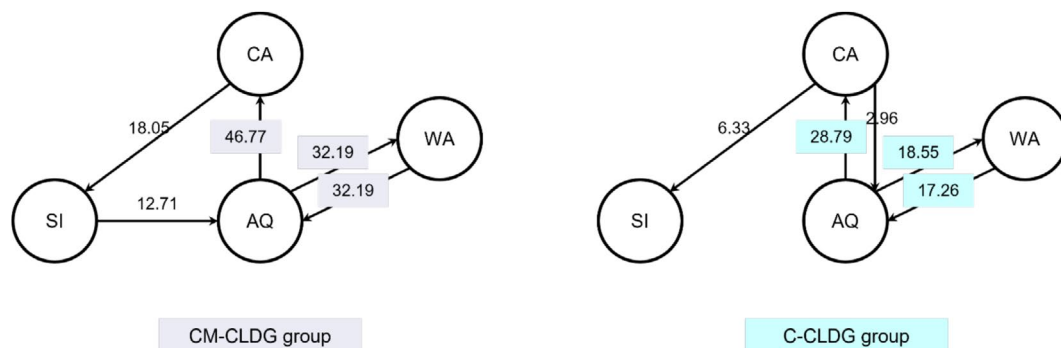


FIGURE 10 | Behavioural transitions related to task engagement through answering.

Figure 10 illustrates students' answer engagement behaviours in both the CM-CLDG and C-CLDG groups. In both groups, transitions from AQ to CA and from AQ to WA (Wrong Answer) were observed, indicating that students consistently engaged in answering tasks regardless of their response accuracy. This suggests that the embedded guidance in both game models effectively supported students in participating in the learning activities. The presence of both correct and incorrect responses across groups reflects a shared tendency to attempt task completion and remain cognitively involved throughout the game-based learning process. These findings highlight the role of task design and in-game scaffolding in sustaining student engagement.

Figure 11 presents the behavioural transitions related to repeated answering in both groups. In the CM-CLDG group, a one-way transition from AQ to CA was identified, showing student progression from response to feedback. In contrast, the C-CLDG group not only exhibited the AQ to CA transition but also a reversed path from CA back to AQ. While this reversed transition appeared less frequently, its presence may suggest that some students repeated the answering process after receiving correct feedback. This pattern could reflect nonstrategic repetition, potentially resulting from uncertainty, accidental actions or limited in-task guidance. Although further data would be needed to confirm the cause, this observation highlights the importance of designing learning environments that support clearer task progression.

Figure 12 presents the behavioural transitions between SI and AQ in both the CM-CLDG and C-CLDG groups. In the CM-CLDG group, a significant transition from SI to AQ was observed, suggesting that students frequently proceeded to answer tasks after receiving contextual information. In this game mode, the concept map served as a guiding framework, requiring students to identify and collect missing elements based on scene cues. This connection between SI and concept map completion may have supported more purposeful task engagement. In contrast, no significant transition from SI to AQ was found in the C-CLDG group. While students accessed the same scene content, the absence of explicit visual scaffolding may have resulted in weaker links between context and task execution. These results highlight the potential of concept map integration to enhance students' responsiveness to scene-based prompts during gameplay.

Figure 13 displays the behavioural transitions related to RK (Review Knowledge) in both gaming modes. In the C-CLDG

group, a significant transition from RK to AQ was observed. This indicates that students sometimes proceeded to answer tasks after reviewing knowledge content. In the CM-CLDG group, although review behaviours were also present, no significant transition from RK to AQ was detected. Given that the CM-CLDG model involved concept maps as a guiding framework, it is possible that students engaged in review for purposes beyond immediate answering, such as identifying concept gaps or updating their understanding through the concept map. The difference in observable transitions between the two groups suggests that the presence or absence of visual scaffolding may influence how students connect review activities with subsequent task responses.

Across Figures 10–13, both groups demonstrated active participation in answering and revising tasks under game-based guidance.

However, the CM-CLDG group exhibited more consistent transitions linked to scene prompts, review behaviours and corrective actions. These patterns suggest that the integrated concept map may have provided clearer guidance, supporting students in maintaining focus and navigating tasks with greater continuity.

5.7 | Student Interviews

Following the teaching experiment, interviews were conducted with six students from the experimental group and six from the control group to collect qualitative insights into their learning experiences. To ensure representativeness, the selection was based on the students' pretest performance. Specifically, three high-achieving and three low-achieving students were

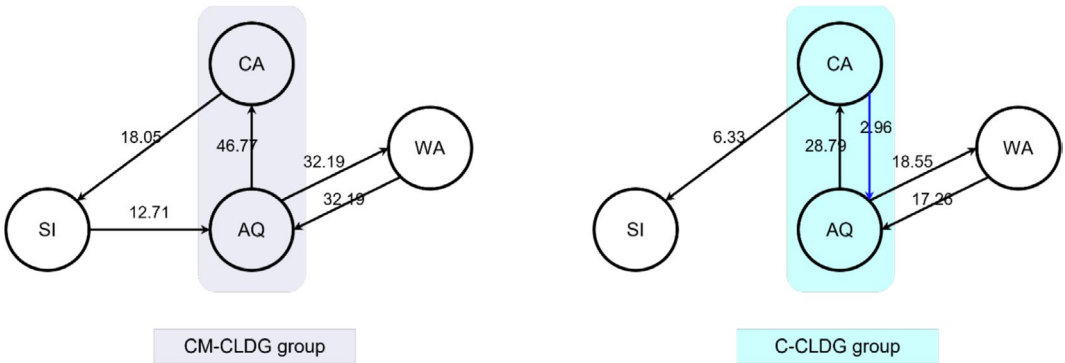


FIGURE 11 | Behavioural transitions between answering and outcome feedback.

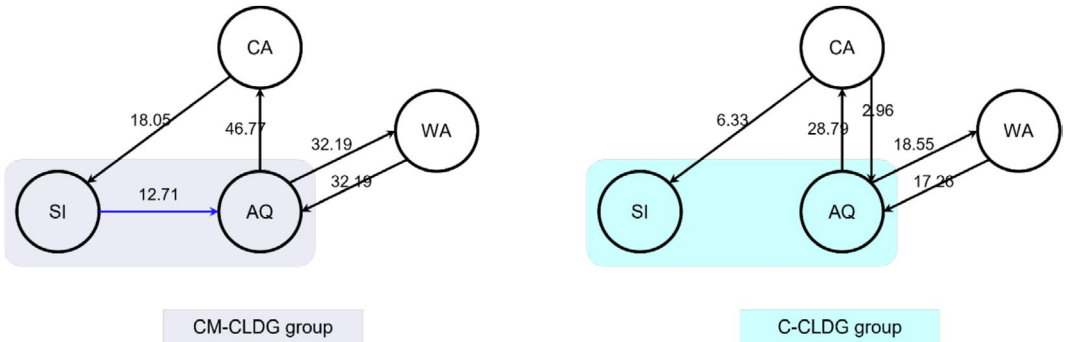


FIGURE 12 | Behavioural transitions from scene information to task engagement.

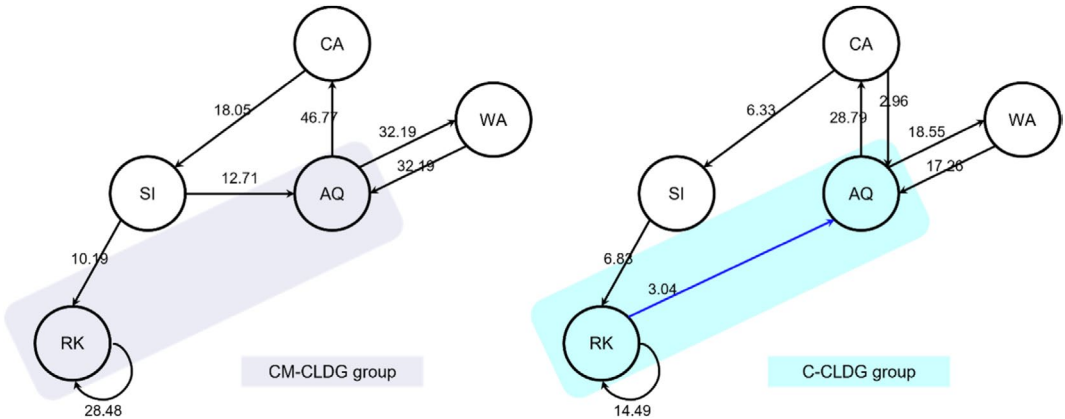


FIGURE 13 | Behavioural transitions involving review and answering actions.

randomly selected from each group. Most participants in both the CM-CLDG and C-CLDG groups expressed positive attitudes toward the contextualised game-based environment. The results also revealed different patterns of experience in the two groups, as summarised in Table 9.

Students in the CM-CLDG group most frequently mentioned enhanced memory of lesson content (12 mentions) and promotion of self-directed learning (six mentions), suggesting that the integration of the concept map supported content retention and encouraged independent learning behaviours. For example, one student noted, 'Compared with static pictures and text, this method helped me much more' (E01), while another explained, 'After organising with the concept map, I knew which parts were the key points' (E02). Others emphasised that combining the concept map with the storyline made the content clearer (E03). Regarding self-directed learning, students pointed out that the digital game environment allowed them to proceed at their own pace: 'In the computer lab, I can learn at my own speed' (E05) and 'Practicing through the game was more fun, and I could re-view until I got it right' (E06).

In contrast, students in the C-CLDG group most commonly reported increased learning motivation (12 mentions) and higher levels of engagement (12 mentions), indicating that the conventional game format, although lacking concept map scaffolding, may have facilitated more immediate and emotionally driven task involvement. One student commented, 'This way of learning is more interesting and makes me want to learn instead of feeling sleepy' (C01), while another added, 'I hope English can continue to be taught in this way, because it feels livelier' (C03). Regarding engagement, students emphasised the interactive and competitive nature of the game: 'Other classes are just sitting and listening to the teacher, but in this class, I could look at the screen and operate by myself; it was much more fun' (C04); 'The competition in the game made it even more engaging' (C05). These reflections illustrate how the two instructional models supported different dimensions of the learning experience: the CM-CLDG model strengthened structured learning and self-regulation, while the C-CLDG model enhanced motivation and active participation.

TABLE 9 | Results of interview-based comparison of learning experiences between the CM-CLDG and C-CLDG groups.

Theme	Mentions	
	CM-CLDG	C-CLDG
Enhanced memory of lesson content	12	6
Improved knowledge application	7	3
Increased learning motivation	7	12
Promotion of self-directed learning	6	2
Greater sense of participation	5	4
Higher level of engagement	5	12
Experience of learning anxiety	1	4

6 | Discussion

6.1 | English Learning Achievement

This study found that students in the CM-CLDG group significantly outperformed those in the C-CLDG group in terms of English learning achievement. The integration of concept maps into the game-based learning environment provided structured scaffolding, which helped students visualise and connect vocabulary and sentence patterns with contextual knowledge (Novak and Cañas 2008). These findings align with prior research demonstrating that concept mapping enhances language learning outcomes by supporting cognitive processing and metacognitive reflection (Chen and Hwang 2020; Hwang and Zou 2020). The embedded visual structure enabled learners to better internalise content, thereby improving their comprehension and retention of the English materials. From the perspective of constructivist learning theory, concept maps act as external representations that support knowledge construction by enabling learners to actively organise and integrate linguistic and content knowledge. This process reduces extraneous cognitive load (Sweller 1994) and enhances schema formation, thereby improving comprehension and retention of the English materials.

6.2 | Science Learning Achievement

Results also showed that the CM-CLDG group had significantly higher science learning achievement than the C-CLDG group. The structured guidance offered by the embedded concept maps supported learners in organising and applying scientific knowledge related to animal habitats. This aligns with previous findings that concept mapping is effective in science education for helping students clarify complex relationships and retain conceptual understanding (Chang et al. 2022; Chiou 2008). By integrating content visualisation directly into the game tasks, the CM-CLDG model effectively facilitated both content mastery and cognitive engagement, fulfilling the dual goals of CLIL instruction. According to dual coding theory (Paivio 1991), combining visual representations (concept maps) with verbal explanations facilitates deeper processing by engaging both visual and verbal channels, thereby strengthening conceptual connections. Moreover, CLIL emphasises the integration of content and language, and concept maps function as scaffolding tools consistent with Vygotsky's sociocultural theory, guiding learners within their zone of proximal development (Vygotsky 1978).

6.3 | Learning Motivation

No significant difference was found in the learning motivation of the CM-CLDG and C-CLDG groups. This may be attributed to students' familiarity with technology-enhanced instruction, as the school had already implemented various digital learning tools in daily teaching (Hwang et al. 2020). Although concept maps offer organisational support, their potential to increase motivation may be offset by reduced immediacy and interactivity (Charsky and Ressler 2011). Another possible explanation lies in the design of the concept mapping tasks. If the tasks

are perceived as overly complex, repetitive or insufficiently integrated into gameplay mechanics, students may regard them as additional burdens rather than engaging challenges. This study suggests that in digital game-based environments, enhancing intrinsic motivation may require more refined designs that effectively combine structured guidance with game-based elements (Charsky and Mims 2008). Future optimisation could involve simplifying concept map structures, incorporating adaptive scaffolding or embedding more interactive features to balance cognitive support with game immersion. According to dual coding theory (Paivio 1991), combining visual representations (concept maps) with verbal explanations facilitates deeper processing by engaging both visual and verbal channels, thereby strengthening conceptual connections.

6.4 | Flow Experience

Students in the C-CLDG group reported significantly higher levels of flow experience compared to those in the CM-CLDG group. While both groups engaged with the same storyline and content, the addition of concept mapping in the CM-CLDG group may have increased cognitive load and interrupted the immersive nature of gameplay (Ke 2008). According to flow theory, a mismatch between task challenge and learner skill may hinder the experience of flow (Csikszentmihalyi 1990). The results indicate that while concept maps promote reflective learning, they may reduce feelings of uninterrupted engagement, suggesting a trade-off between structured learning and game immersion. From the perspective of cognitive load theory (Sweller 1994), the embedded concept mapping tasks may have simultaneously increased intrinsic and extraneous load, as learners needed to process both linguistic and scientific knowledge while interacting with the game environment. This divided attention reduced the cognitive resources available for immersion, thereby reducing the flow experience. At first glance, this finding seems to contradict Hwang et al. (2013), who reported that concept maps embedded in digital games reduced cognitive load. One possible explanation lies in the differences in learning context and task design. In our study, concept maps required the integration of cross-disciplinary content (English and science) and the active use of sentence patterns and vocabulary, which may have increased intrinsic load. In contrast, Hwang et al.'s study primarily focused on content knowledge, where concept maps served more as a structural aid to reduce extraneous load. Thus, although both studies implemented embedded concept maps, the cognitive effects varied depending on the complexity of the learning tasks and the learners' prior familiarity with bilingual content integration. Taken together, these findings highlight that embedded concept mapping can both support deep learning and increase cognitive demands. Designers must therefore carefully consider task complexity and learner readiness when integrating concept maps into game-based environments to balance reflective learning with immersive flow.

6.5 | Learning Anxiety

Analysis showed no significant difference in learning anxiety between the CM-CLDG and C-CLDG groups. This may be due

to the students' previous exposure to digital learning, which could have normalised the game-based environment and reduced potential anxiety triggers. Although prior research has found that concept maps may alleviate language anxiety by providing clearer structures and reducing ambiguity (Chen and Hwang 2020), the already moderate anxiety levels observed in this study may have limited the potential for measurable differences. Future studies may consider applying similar interventions in higher-anxiety contexts to better assess the affective benefits of concept mapping. Krashen's affective filter hypothesis (1982) suggests that anxiety influences language acquisition by either facilitating or impeding input processing. In this study, the relatively low affective filter across both groups may have reduced the observable impact of concept mapping on anxiety reduction.

6.6 | Learning Behaviours

Sequential analysis revealed that both groups actively engaged in answering tasks, as evidenced by consistent AQ → CA transitions. This indicates that the built-in game mechanisms effectively promoted learner participation. However, the C-CLDG group also exhibited a reversed CA → AQ path, suggesting nonstrategic repetition possibly due to confusion or interface limitations. A unique SI (Scene Info) → AQ transition was found only in the CM-CLDG group, indicating that students effectively used scene prompts to complete concept maps and guide task responses. Additionally, the C-CLDG group showed a significant RK → AQ transition, while the CM-CLDG group did not, possibly because learners in the CM-CLDG model engaged in more reflective or delayed responses via concept mapping (Chang et al. 2022). Mechanistically, the SI → AQ pathway in the CM-CLDG condition reflects a selection–organisation–integration cycle: learners first activate relevant concepts from scene cues (selection), then place them into node–link structures (organisation), and finally articulate relations among concepts (integration). This generative processing externalises tacit links as propositions and cross-links, creating richer retrieval cues and facilitating schema construction, which are core processes of knowledge internalisation and later transfer. These results suggest that concept maps enhance structured and strategic learning behaviours during gameplay. From the perspective of metacognitive theory (Flavell 1979), the CM-CLDG design promoted planning and monitoring behaviours, as evidenced by the use of SI before answering. Consistent with cognitive load theory, concept mapping reallocates effort from extraneous to germane load by channelling attention to relation-building rather than trial-and-error responding; this explains the reduction of CA → AQ 'guess-check' loops in CM-CLDG and the prominence of SI → AQ planning moves. In contrast, the RK → AQ pattern in C-CLDG suggests surface rehearsal; without map-based structuring, learners rely on short-term review before answering, which yields participation but weaker integrative encoding. In sum, the observed behavioural sequences indicate that concept maps scaffold the micro-processes (planning → mapping → justifying) through which gameplay events are converted into coherent conceptual networks, thereby deepening internalisation beyond immediate task completion. This indicates that concept maps functioned as metacognitive

scaffolds, encouraging learners to adopt more strategic and reflective approaches during gameplay.

6.7 | Student Interviews

Interview data provided further insight into students' learning experiences. Students in the CM-CLDG group most frequently mentioned improved memory retention and increased self-directed learning, indicating that the concept maps helped them organise and internalise the learning content (Novak and Cañas 2008). In contrast, students in the C-CLDG group more often reported heightened motivation and engagement, suggesting that the streamlined gameplay promoted emotional involvement (Gee 2003; van Eck 2006). Interestingly, more students in the C-CLDG group mentioned experiencing learning anxiety, which may reflect the lack of structural guidance. These findings support the conclusion that concept maps not only enhance cognitive processing but also contribute to more autonomous learning behaviours in a CLIL game-based context. This aligns with Zimmerman's self-regulated learning theory (2000), which emphasises that tools fostering planning, monitoring and reflection, such as concept maps, can strengthen learners' autonomy and self-directed strategies in CLIL contexts.

7 | Conclusions

This study contributes to the growing body of research on CLIL by proposing and evaluating a novel instructional model that embeds concept maps into DGBL environments. Through triangulated evidence drawn from learning achievement, learner perceptions and behavioural patterns, the study demonstrates that the CM-CLDG approach effectively enhances both English and science learning outcomes among elementary school students. Furthermore, the integration of concept maps into gameplay was shown to promote strategic and reflective learning behaviours, addressing the cognitive challenges often observed in unstructured game-based learning. For teachers, this model provides a practical pathway to embed scaffolding strategies into game-based activities without diminishing student engagement. By integrating concept mapping into digital games, teachers can guide students to organise knowledge systematically, monitor their own progress and reflect on their learning, which can be directly incorporated into classroom practice to support bilingual instruction. These findings offer practical implications for instructional designers and educators seeking to implement structured yet engaging bilingual learning experiences.

Several limitations must be acknowledged. First, the sample size was relatively small and limited to a single school, which may restrict the generalisability of the findings. Second, while the study controlled for prior knowledge using pretests, other learner characteristics such as learning styles, digital literacy or prior exposure to game-based environments were not assessed. Third, the intervention period was short-term; thus, the study did not evaluate the long-term retention of content knowledge or sustained motivation effects. Finally, while behavioural sequences were analysed using GSEQ and were complemented

by student interviews, the in-depth exploration of learners' metacognitive processes during gameplay warrants further investigation.

Future research could expand this study by involving a larger and more diverse sample to improve generalisability. Longitudinal designs are recommended to assess the enduring effects of the CM-CLDG approach on knowledge retention and motivation. In addition, further investigations should examine how learners' individual differences, such as prior gaming experience or sensitivity to cognitive load, may influence the effectiveness of concept map-embedded digital gaming environments. Studies may also explore the use of adaptive scaffolding mechanisms to help manage cognitive demands during gameplay. Moreover, the CM-CLDG model has potential for transferability to other subject domains beyond English and science, such as social studies, mathematics or health education, where complex concepts and disciplinary language need to be integrated. Applying this model across disciplines could provide further evidence of its generalisability and practical utility in bilingual and CLIL contexts. Finally, integrating behavioural data with richer qualitative methods, such as think-aloud protocols or screen-capture analysis, could provide deeper insights into students' reasoning and metacognitive processes throughout the learning experience.

Author Contributions

Shu-Fang Hsieh: conceptualization, methodology, investigation, data curation, writing – original draft, formal analysis. **Gwo-Jen Hwang:** conceptualization, software, validation, investigation, supervision, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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